Augmented Reality Workshop

Animated Dancing Bitmoji

Helper Document / Snap Chat - Lens Studio Project

Last Updated: January 2025

Project insight

This lens studio project aims to create an augmented reality experience for snapchat with an animated bitmoji. When the effect is loaded, and while looking through the rear camera, an bitmoji of the account holder will be placed.

If tap and hold with a finger on the screen, the bitmoji character will start dancing. And when let go, the dance will be stopped. This will be progressed with multiple custom animations that you create as a part of this project.

We will be setting up the account, gather all the downloadable resources, prepare some assets. install and setup lens studio to start a new project.

01

Downloading project files

Link

Download the project files for animated dancing bitmoji from github repository. Click on the green “Code” button to download.

02

Snapchat account

Link

Go to the official Snapchat website and create a Snapchat account, then login.

03

Download lens studio

Link

Downloading the latest lens studio available [5.6 or later]

04

Download the bitmoji fbx from snapchat

Link

Click this link to download the Bitmoji FBX for animation purposes.

05

Installing the lens studio

Double click on the downloaded executable to install the lens studio. To proceed, click on next/ok when asked.

06

Mixamo account

Link

Go to Mixamo and create an account. You can use your google account to sign up as well or login with your adobe account.

07

Mixamo upload

Upload your “Animated Bitmoji 3D FBX” zip file from snapchat to mixamo. Wait for maxiamo to process the rig. If asked, match the joints in the rig.

08

Mixamo animations

- Choose a custom animation for each of “Idle”, “Wave”, “Dance” & “Stumble”.
- Download all the animations* as FBX file. Use a 30fps format.
- Save the filenames of the fbx files as the name of the animation itself - Idle, Wave, Dance & Stumble.
- Keep a note on how many frames each animation has.

* You can choose any custom animations you like for this project, but for the sake of uniformity of this project, we will be going with Idle, Wave, Dance & Stumble animations. These default animations are provided in the downloaded project files.

09

Open lens studio 5.6+

Open the lens studio. Login to your snapchat account.

10

New project

Create a blank project, not templates. Save the project with “ctrl+s” / “cmd+s” to you desired location.

We'll add the necessary assets and begin configuring the project.

01 Adding animations

- On the asset browser, drag in all the 04 required animations - Idle, Wave, Dance & Stumble.
- While adding, on the import pop-up, make sure to uncheck “Use Legacy Importer”.
- Optionally, after adding the animation assets, you can organize them in an “Anim” folder in the assets browser.

02 Adding assets from library

- Open the assets library [found above the “scene hierarchy”].
- Search for “Bitmoji 3D”, “Animation State Manager” & “World Object Controller” components and add them.

03 Create a new scene object

- In the scene hierarchy, click on the “+” button. Add a new scene object from General > Scene Object or you can search for it.
- Rename this object to “Bitmoji”.

04 Adding the component

- On the bitmoji object, click on the “Add Component” button on the inspector panel and add a “Bitmoji 3D” component.
- Make sure to enable* “Adapt to Mixamo”.

*Do this step if you have imported animations from mixamo, otherwise uncheck the Adapt to Mixamo option.

05 Camera settings

- Switch the simulator view to rear camera.
- Add a “Device Tracking” component to the camera object.

We will be adding all the animation clips to the bitmoji object. Animation clips are helpful in create clips from our custom animation fbx assets.

01

Animation player

- On the Bitmoji object, add an “Animation Player” component.
- On the animation player component, click on “Add clip” and choose the first animation: Idle.
- Repeat the process for the rest of the clips: Wave, Dance & Stumble.
- Rename* the clip to the filenames

* You can name the clips anything, but for the sake of simplicity and easy organization, we are going with the filenames of the provided animation objects.

02

Idle clip

- Animation Asset: Idle
- Name: Idle
- Begin: 0
- End: 210
- Playback Mode: Loop

03

Wave clip

- Animation Asset: Wave
- Name: Wave
- Begin: 0
- End: 38
- Playback Mode: Single

04 Dance clip

- Animation Asset: Idle
- Name: Idle
- Begin: 0
- End: 247
- Playback Mode: Loop

05 Stumble clip

- Animation Asset: Stumble
- Name: Stumble
- Begin: 0
- End: 168
- Playback Mode: Single

06 Animation state manager

- Add a new animation state manager in the assets browser using the “+” button.
- Drag the animation state manager from assets browser to the scene hierarchy.
- Select the animation state manager and add the “Bitmoji” object as the “Animation Player” in the inspector panel.
- Enable “Create From Clips” to use the animation from the bitmoji object.

We'll configure the animation state manager to use clips like Idle, Wave, Dance, and Stumble, allowing for interactive animations based on specific conditions and triggers.

01 Animation state manager

- Component > Parameters > Edit > Enable
  - Click on “Add Value”
    - Name: isDancing
    - Type: bool
- Component > Transitions > Edit > Enable
  - Click on “Add Value”
    - From: Entry
    - To: Specified State
      - Name: Idle

02 Behavior script {Touch Start}

- On the scene hierarchy, click on the “+” button, search for “Behavior Script” and add it to the scene.
- Rename to “Behavior Script - Touch Start”
- Change the behavior script settings to
  - Trigger: Touch Event
  - Event Type: Touch Start
  - Response Type: call Object API
  - Target
    - Target Type: Script or Component
    - Component: Animation State Manager (from the scene hierarchy)
  - Settings
    - Function Name: setParameter*
      - Argument 1
        - Value: String
        - String Value: isDancing
      - Argument 2
        - Value Type: bool
        - Bool Value: TRUE

* setParameter is an inbuilt function of the behavior script and can be found in the description of the behavior object in the assets browser. This behavior script will check if the parameter is met or not, and sends a bool value.

03

Animation state manager

On the transition slots of the animation state manager object, click on the “Add Value” button to add different transition states.

- From
  - Name: Idle
- To
  - Name: Dance
  - Duration: 0.2
- Condition
  - Name: isDancing
  - Type: Bool
  - Function*: is True

*State machine will check if this condition is met, then execute this transition from idle to dancing.

04

Behavior script {Touch End}

- Duplicate the “Behavior Script - Touch Start”
- Rename to “Behavior Script - Touch End”
- Change the behavior script settings to
  - Trigger: Touch Event
  - Event Type*: Touch End
  - Response Type: call Object API
  - Target
    - Target Type: Animation State Manager (from the scene hierarchy)
  - Settings
    - Function Name: setParameter
      - Argument 1
        - Value: String
        - String Value: isDancing
      - Argument 2
        - Value Type: bool
        - Bool Value*: FALSE

*The only change here is event type and argument 2’s bool value. This make sures to revert back to idle animation from dancing when the touch is stopped. We can add in-between animations to like from dancing to stumble to idle.

05

Animation state manager

On the transition slots of the animation state manager object, click on the “Add Value” button to add different transition states.

- From
  - Name: Dance
- To
  - Name: Stumble
  - Duration: 0.0
- Condition
  - Name: isDancing
  - Type: Bool
  - Function*: is False

*State machine will check if this condition is met, then execute this transition from dancing to stumble.

06

Animation state manager

On the transition slots of the animation state manager object, click on the “Add Value” button to add different transition states.

- From
  - Name: Stumble
- To
  - Name: Idle
  - Duration: 0.1
- Has Exit Time: TRUE
  - Exit Time: 0.8
- Condition
  - Name: isDancing
  - Type: Bool
  - Function*: is False

*State machine will check if this condition is met, then execute this transition from stumble to idle.

07

Animation state manager

Scroll to the top and add a new parameter in the parameters section.

- Name: waveTrigger
- Type: Trigger

08

Animation state manager

On the transition slots of the animation state manager object, click on the “Add Value” button to add different transition states.

- From
  - Name: Idle
- To
  - Name: Wave
  - Duration: 0.1
- Condition
  - Name: waveTrigger
  - Type: Trigger

*State machine will check if we are looking at the bitmoji, then execute this transition from idle to wave.

09

Animation state manager

On the transition slots of the animation state manager object, click on the “Add Value” button to add different transition states.

- From
  - Name: Wave
- To
  - Name: Idle
  - Duration: 0.15
- Has Exit Time: TRUE
  - Exit Time: 1.0
- Condition
  - Name: waveTrigger
  - Type: Trigger

*State machine will check if we are not looking at the bitmoji, then execute this transition from wave to idle.

10

Behavior script {Looking At}

- Duplicate the “Behavior Script - Touch Start”
- Rename to “Behavior Script - Looking At”
- Change the behavior script* settings to
    - Trigger: Looking At
        - Looking Object: Camera Object
        - Looking Target: Bitmoji
    - Angle: 20.0
    - Settings
        - Function Name: setTrigger
            - Argument 1
                - Value: String
                - String Value: waveTrigger*
            - Argument 2
                - Value Type: None

* setTrigger is an inbuilt function of the behavior script and can be found in the description of the behavior object in the assets browser. This behavior script allows the user to check if the camera is pointing at the bitmoji object or not, and sends a trigger.

We'll enhance the interactivity and realism.

01

World object controller

- Once you add the “world object controller” object into the scene, replace the “Sphere [REPLACE ME]” object with the “Bitmoji” object.
- Make sure to add the bitmoji object in same level as the sphere[replace me] object, to properly use the world object controller.

02

Light

Select the light object under, Lighting > Light and enable* “Shadows”

*By enabling the shadows from the source light, the World Object Controller will catch the shadows from casted from the Bitmoji object using the “Ground Grid” object.

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We'll test the effect on our mobile device and upload it to Snapchat for global use.

01 Preview lens

- On the top right of the screen near the publish button, you can choose to pair with your mobile device using your snapchat account.
- Select “Pair new Snapchat Account” and a SnapQR* will be shown.
- Open your snapchat and point the camera to the SnapQR.
- You will receive the prompt to continue the pairing process.
- Once paired the lens will send to your device.

* SnapQR can only be scanned using a snapchat camera. Normal QR code readers won't work with SnapQR.

02 Automating lens previews

- Click on Lens Studio on top left and select “Preference”.
- Send to Device > Send on Project Save: Snapchat Only.

03 Publish

- On the top-right, select ”Publish” to open the project settings window.
- You can add or change the Lens Name, Platforms, Previews and Icons.
- Once customized, you can again select “Publish”.
- That will guide you to the snapchat website. Follow the instructions and your effect will be published in a while.

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